AIRINDEX

UAM Market Readiness Index

Methodology Brief

Version 1.2.3 · March 2026 · Vertical Data Group, LLC

DOCUMENT OVERVIEW

This document describes the data sources, classification engine, scoring framework, and validation methodology underlying the AirIndex UAM Market Readiness Index — a systematic, data-driven assessment of U.S. city readiness for commercial eVTOL operations.

21	4	7	~89%
U.S. Markets Scored	Primary Data Sources	Scoring Factors	Classification Accuracy

1 Purpose and Scope

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1 Purpose and Scope

AirIndex scores U.S. cities on their readiness for commercial eVTOL (electric vertical takeoff and landing) operations using a systematic, data-driven methodology — not editorial judgment. The index is designed to give operators, city planners, investors, airport authorities, and policy researchers a consistent, auditable framework for comparing market readiness across geographies.

The index currently covers 21 U.S. markets, scored on a 0–100 scale across seven weighted factors. Scores are updated automatically when qualifying regulatory, legislative, or operator events are detected by the ingestion pipeline, with human review triggered for medium and low-confidence classifications.

> **What the index does not measure:** AirIndex does not predict commercialization timelines, assess operator financial viability, or evaluate aviation safety outcomes. Scores reflect market readiness conditions — not guarantees of eVTOL deployment.

2 Scoring Framework

Each city is scored across seven factors. Six factors are binary (present/absent for full points). One factor — State Legislation — uses a graduated three-tier model introduced in v1.2.1. The maximum possible score is 100.

SCORING FACTOR	POINTS	RATIONALE
Active Pilot Program	20 pts	Presence of a sanctioned UAM/eVTOL pilot program — the strongest single indicator of near-term commercial readiness.
Approved Vertiport	20 pts	At least one vertiport has received regulatory approval, indicating physical infrastructure commitment.
Active Operator Presence	15 pts	A certified eVTOL operator has announced or commenced operations in the market.

Vertiport Zoning	**15 pts**	Municipal or county zoning frameworks explicitly accommodate vertiport infrastructure.
Regulatory Posture	**10 pts**	The local regulatory environment is demonstrably favorable to UAM operations based on FAA and municipal signals.
State Legislation	**10 pts**	Graduated three-tier model: **enacted** (10 pts) — state has passed UAM/eVTOL legislation; **actively_moving** (5 pts) — bills in late legislative stages (committee passage, floor votes, 2nd reading); **none** (0 pts). Source: LegiScan primary legislative data.
LAANC Coverage	**10 pts**	Low Altitude Authorization and Notification Capability (LAANC) is active in the market area, enabling automated airspace access.

Score Tiers

TIER	RANGE	DESCRIPTION
ADVANCED	75 – 100	Market has most or all infrastructure, regulatory, and operator requirements in place. Commercial UAM operations are imminent or active.
MODERATE	50 – 74	Significant progress across multiple factors. Key pieces are in place but gaps remain — typically missing infrastructure or operator commitment.
EARLY	30 – 49	Some foundational elements present. Regulatory posture may be favorable but physical infrastructure and operator activity are limited.
NASCENT	0 – 29	Minimal UAM readiness. Market may have LAANC coverage or early regulatory signals but lacks substantive infrastructure or operator engagement.

Current Market Snapshot — March 2026

The index currently covers 21 U.S. markets with an average score of 39 (EARLY tier). Three markets have achieved perfect scores: Los Angeles (100), Dallas (100), and Miami (100).

Miami was most recently updated on March 13, 2026, following automated detection of an FAA Federal Register filing confirming Archer Aviation's White House AAM Pilot Program selection. Chicago moved from 30 to 35 (EARLY tier) on March 14, 2026, following graduated legislation scoring recognizing IL HB3190.

3 Data Sources

AirIndex ingests data from four primary sources. Each source feeds a distinct category of scoring signal. As of March 2026, the active dataset contains **1,403** ingested records across all four sources.

Federal Register

6.0% · 84 records

FAA rulings, NOPRMs, Part 135 applications, Special Federal Aviation Regulations (SFARs), and airspace notices. The most authoritative government source for regulatory posture and pilot program signals. Active lookback window: 730 days (2 years). 12 search terms including "eVTOL Integration Pilot Program" and FAA docket monitoring. Updated: daily cron ingestion.

Federal Register coverage expanded from 5 to 84 records in March 2026 via pagination fix, double-filter removal, and search term expansion (v1.1). Additional search terms for FAA eVTOL/AAM Integration Pilot Program added March 14, 2026 (v1.2).

LegiScan

28.4% · 399 records

State-level UAM and UAS legislation across all 50 states — bills, resolutions, and committee actions. Primary source for the State Legislation scoring factor, including graduated status assessment (enacted vs. actively_moving vs. none). Coverage filters applied to UAM-relevant legislative activity.

SEC EDGAR

5.6% · 79 records

Operator filings from publicly traded eVTOL companies including Joby Aviation and Archer Aviation (8-Ks, 10-Qs, and material event disclosures). Used to detect operator market expansion, partnership announcements, and certification milestones.

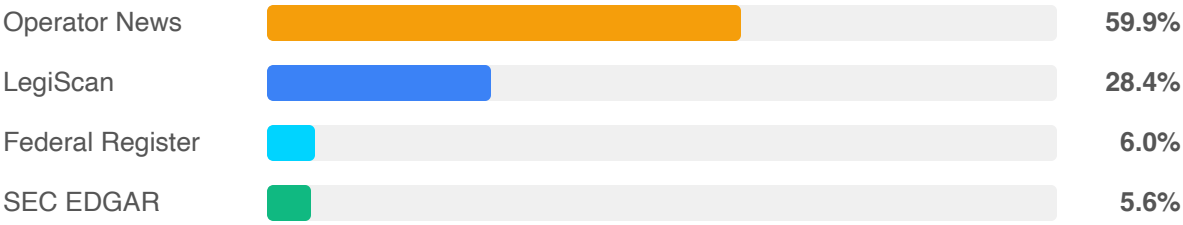
Operator News

59.9% · 841 records

Curated operator announcements, market expansions, and industry milestones from primary sources including company press releases and aviation trade publications. Selection criteria: announcements must reference a specific market, regulatory action, or operational milestone. General financial coverage and stock analysis are excluded.

> *Operator News is a watch list and intelligence signal source. Per the Score Promotion Policy (Section 6), news coverage alone does not qualify for automatic score promotion on primary-source-required factors.*

Source Distribution



4 Classification Engine

Raw ingested records are processed by an AI classification engine (Claude Haiku 4.5) that assigns each record an event type, confidence level, affected city tags, and relevant scoring factors. This classification drives scoring override recommendations — the mechanism by which new information updates city scores.

Confidence Tiering

Every classification result includes a confidence field with one of three values. The confidence level determines how the record moves through the pipeline:

CONFIDENCE	VOLUME	BEHAVIOR	HUMAN REVIEW
HIGH	84%	Classification flows to auto-reviewer. High-confidence overrides evaluated for automatic score promotion (subject to source quality requirements).	Only if auto-reviewer rejects
MEDIUM	12%	Flagged for human review queue. Not promoted to scoring automatically.	Required before any score change
NEEDS REVIEW	4%	Manual intervention required. Record held pending human classification decision.	Always required

Classifier Prompt Version History

The classification prompt is iteratively refined based on accuracy audits, false negative analysis, and pipeline performance data. Every prompt version is tracked with a commit hash, enabling precise reproducibility of any historical classification.

VERSION	DATE	CHANGES
v1	Feb 28, 2026	Initial classifier. Generic factor keys, basic event type taxonomy, single-pass classification.

v2	Mar 4, 2026	Field mapping standardization — aligned classifier output keys to scoring engine fields (`activePilotProgram` → `hasActivePilotProgram`). Content enrichment added (SEC filings 2K chars, operator news 1.5K chars).
v3	Mar 11, 2026	Confidence calibration. Established ~80% accuracy baseline across 34-record retrospective audit.
v4	Mar 13, 2026	Stock-framed false negative fix (3 few-shot examples). Metro area geographic mappings (17 areas). Federal vs. state factor distinction rules. Accuracy improved to ~88–90%.
v5	Mar 14, 2026	DOT/FAA pilot program few-shot examples for government programs wrapped in stock commentary. URL deduplication improved — MarketScreener domain-level normalization and tracking parameter stripping.

Audit Trail

Every classification and scoring decision is fully logged. The system maintains three persistent audit tables:

- **ClassificationResult** — Stores every classification with raw response, confidence level, model used, prompt version, and factors JSON.
- **ScoringOverride** — Tracks every override with confidence, reason, source record ID, applied timestamp, and superseded timestamp.
- **AutoReviewResult** — Logs every auto-review decision with reasoning, confidence, application status, and source content.

> Every score change is traceable to a specific source record, classification result, and override decision — including the timestamp, confidence level, and reasoning. This audit trail is maintained indefinitely and available for methodology review upon request.

Signal Distribution — March 2026

EVENT TYPE	COUNT	DESCRIPTION
faa_certification_milestone	31	FAA type certificate actions, conforming aircraft approvals, test flight authorizations

operator_market_expansion	25	Operator announcements of new market entry, partnerships, or route launches
state_legislation_signed	3	State-level UAM/eVTOL legislation enacted
faa_corridor_filing	1	FAA corridor authorization or airspace notice
not_relevant	146	Correctly filtered noise — stock analysis, insider trading disclosures, unrelated regulatory filings

5 Accuracy and Validation

Two retrospective accuracy audits have been conducted in March 2026, establishing a documented improvement arc across prompt versions. Each audit sampled classifications across all three confidence buckets. Records were manually reviewed against source content to verify correct event type assignment, city tagging, and scoring factor mapping.

v1.1 Accuracy Audit — March 13, 2026 (Prompt v3)

BUCKET	SAMPLE	CORRECT	QUESTIONABLE	WRONG	ACCURACY
HIGH	20	15	4	1	75 – 95%
MEDIUM	10	7	0	2	70 – 80%
NEEDS REVIEW	4	2	1	1	50 – 75%
OVERALL	**34**	**24**		4	**~80%**

v1.2 Accuracy Audit — March 14, 2026 (Prompt v4)

BUCKET	SAMPLE	CORRECT	QUESTIONABLE	WRONG	ACCURACY
HIGH	20	19	1	0	**~95%**
MEDIUM	10	8	1	1	~85%
NEEDS REVIEW	5	3	1	1	~75%
OVERALL	**35**	**30**		2	**~88–90%**

Improvement Analysis

The v1.2 accuracy improvement from ~80% to ~88–90% is attributable to four targeted prompt fixes deployed between audits:

FIX AREA	v1.1 STATUS	v1.2 STATUS	RESULT
Stock-framed false negatives	Active failure mode	**Resolved**	7/7 stock-framed headlines correctly filtered in HIGH bucket
Bay Area metro mapping	Inconsistent	**Resolved**	6/6 Bay Area locations correctly mapped to san_francisco
Federal vs. state distinction	**Occasional mismatch**	**Resolved**	Arizona correctly mapped to Phoenix, federal programs distinguished
URL deduplication	Duplicates present	**Resolved**	MarketScreener domain-level normalization and tracking parameter stripping shipped in v1.2.2. Full cross-domain dedup remains a v1.3 target.

Interpretation: The system's primary failure mode remains conservative — it drops real signals rather than promoting false ones. For a scoring system where credibility is the product, this is the correct failure mode. Scores are more likely to be understated than overstated. The DOT "Future of Flight" pilot program edge case was resolved in v1.2.2 (prompt v5) with targeted few-shot examples for government programs wrapped in stock commentary.

6 Score Promotion Policy

Established March 2026. Score changes require source documentation that meets a two-tier quality hierarchy. This policy is enforced automatically in the classification pipeline and auto-reviewer.

Two-Source Hierarchy

SOURCE TIER	EXAMPLES	ELIGIBLE ACTIONS
Primary / Official	Federal Register filings, FAA documentation, DOT notices, LegiScan legislative records, state legislature records, SEC EDGAR filings, official .gov sources	Automatic score promotion at high confidence. Full scoring factor changes including hasActivePilotProgram, hasVertiportZoning, stateLegislationStatus.
Secondary / News	Credible local journalism (WRAL, NY Post), aviation trade publications, operator press releases, industry aggregators	Watch list trigger and elevated activity signal **only**. Does not qualify for automatic score promotion on primary-source-required factors. Suitable for activeOperators and regulatoryPosture changes.

Factors Requiring Primary Source Documentation

FACTOR	WEIGHT	REQUIRED SOURCE	RATIONALE
hasActivePilotProgram	20 pts	Federal Register filing or direct FAA documentation	20-point weight with significant tier implications warrants primary source confirmation
hasVertiportZoning	15 pts	Official city/county zoning records or	Zoning changes require municipal documentation

		council minutes	
stateLegislationStatus	10 pts	LegiScan or official state legislature records	Legislative status must be traceable to bill records

> **Policy rationale:** Multi-source news coverage of FAA programs constitutes strong signal and appropriately triggers watch list monitoring. However, pilot program factor changes carry 20-point weight and can produce tier-changing score movements (e.g., Charlotte 15→35, New York 70→90). Score changes of this magnitude require primary source documentation on the audit trail. News coverage routes to watch list; Federal Register filings route to scoring.

7 Market Watch List

Launched March 14, 2026. The Market Watch List is an automated, pipeline-driven intelligence layer that identifies markets showing signals that haven't fully moved their scores yet. Watch list status is not a score change — it is early intelligence.

Trigger Types

TRIGGER	CRITERIA	EXAMPLE
LEGISLATION	stateLegislationStatus === "actively_moving"	Phoenix (AZ SB1827/SB1819/SB1826), Columbus (OH HR304/HB251), Chicago (IL HB3190)
PIPELINE_OVERRIDE	Unresolved high/medium confidence scoring overrides pending review	Los Angeles, Miami, Houston — pending operator or regulatory overrides
ELEVATED_ACTIVITY	3+ relevant classifications in 30 days with no score movement	Charlotte (NC DOT + FAA AAM pilot program news), New York (FAA air taxi test program news)

Watch List Lifecycle

Entries are evaluated daily as part of the 06:00 UTC cron cycle. Entries auto-resolve when their trigger conditions no longer qualify — either because the underlying signal resolved, the score moved to reflect the change, or legislation was enacted. Resolution reasons are logged for audit purposes.

Current state (March 14, 2026): 15 active entries across 10 markets. Includes 2 markets (Charlotte, New York) with elevated activity from news coverage of FAA eVTOL/AAM Integration Pilot Program participant selections — pending Federal Register confirmation per Score Promotion Policy.

8 Score Updates and Versioning

Scores update when the classification pipeline surfaces a high-confidence override from a primary source that the auto-reviewer promotes to active status. The pipeline runs on a daily cron schedule (06:00 UTC). All score changes are logged with source, timestamp, and reasoning.

Score Change Example — Chicago, March 14, 2026

Market	Chicago, Illinois
Previous Score	30 (EARLY tier)
Updated Score	35 (EARLY tier)
Trigger	LegiScan: IL HB3190 (AERONAUTICS-UNMANNED AIRCRAFT) placed on calendar for 2nd reading
Factor Updated	stateLegislationStatus: none → actively_moving (+5 pts)
Classification	Graduated legislation scoring path (v1.2.1) — LegiScan primary source
Timestamp	March 14, 2026 (manual seed data update, clean legislative audit trail)

Methodology Version Log

Changes to the scoring methodology, data sources, or classification logic are versioned and documented. The current version is 1.2.3.

VERSION	DATE	CHANGES
v1.0	March 2026	Initial methodology. 21 markets, 7 scoring factors, 4 data sources. Federal Register integration active with 90-day lookback and 20 results per term.

v1.1	March 13, 2026	Federal Register fix: pagination added (up to 500 results per term), double-filter removed, new search terms (SFAR, "powered lift"), lookback extended to 730 days. Records grew from 5 to 84. Retrospective accuracy audit completed (34 records, ~80% accuracy).
v1.2	March 14, 2026	Classification prompt v4: stock-framed false negative fix, Bay Area metro geographic mappings (17 areas), federal vs. state factor distinction. Accuracy improved from ~80% to ~88–90% (35-record audit). Two-source hierarchy policy established for score promotion. Federal Register search terms expanded with FAA eVTOL/AAM Integration Pilot Program docket monitoring.
v1.2.1	March 14, 2026	Graduated legislation scoring: State Legislation factor converted from binary (enacted/none) to three-tier model (enacted 10 pts / actively_moving 5 pts / none 0 pts). Based on LegiScan data analysis across 398 records and 49 states. Phoenix +5 (AZ SB1827), Columbus +5 (OH HR304), Chicago +5 (IL HB3190). Market Watch List launched — automated pipeline-driven intelligence layer with 3 trigger types (LEGISLATION, PIPELINE_OVERRIDE, ELEVATED_ACTIVITY). 15 active entries at launch.
v1.2.2	March 14, 2026	Prompt v5: Stock filter tuning for articles containing both regulatory news and stock commentary (DOT/FAA pilot program few-shot examples added). URL deduplication improved — MarketScreener domain-level normalization and tracking parameter stripping.
v1.2.3	March 17, 2026	Confidence cap for news-sourced overrides on primary-source-required fields (hasActivePilotProgram, hasVertiportZoning, stateLegislationStatus). Classifier confidence is now evaluated independently of source quality tier — high-confidence news classifications route to watch list only, not auto-promotion queue. San Francisco score corrected from 75 to 60: Joby Bay Area demo flight is a watch list signal, not pilot program confirmation.

9 Limitations and Ongoing Development

Transparency about current limitations is central to AirIndex's methodology philosophy. The following limitations are acknowledged and tracked with resolution status.

Operator News Manual Curation Dependency

V2

The largest data source by volume (59.9%) relies on curated RSS feeds. While selection criteria are documented, the source is more dependent on active curation than fully automated sources. Full automation of this source is a V2 priority.

Classification Feedback Loop

V2/V3

Manual review decisions (approve/reject overrides) are stored in AutoReviewResult but are not yet used to fine-tune classification prompts automatically. This feedback loop will be closed in V2/V3 when sufficient decisions have accumulated.

False Negative Rate on Stock-Framed Headlines

RESOLVED v1.2.2

High-confidence classifications occasionally dropped real regulatory signals when articles framed underlying events in financial rather than regulatory terms. Resolved in v1.2 prompt update — 7/7 stock-framed test cases correctly filtered. DOT "Future of Flight" edge case resolved in v1.2.2 with targeted few-shot examples for government programs wrapped in stock commentary.

News-Sourced Override Auto-Promotion

RESOLVED v1.2.3

High-confidence classifications from news sources could bypass the two-source hierarchy and auto-promote score changes on primary-source-required factors (hasActivePilotProgram, hasVertiportZoning, stateLegislationStatus). Resolved in v1.2.3 with a confidence cap that evaluates classifier confidence independently of source quality tier. News-sourced overrides on restricted fields are capped at medium confidence regardless of classifier output, routing them to the watch list instead of the auto-promotion queue. San Francisco score corrected from 75 to 60 as a result.

Metropolitan Area Mapping

RESOLVED v1.2

Geographic city tagging previously did not fully account for metropolitan areas (e.g., Oakland → san_francisco). Resolved in v1.2 with 17 metro area geographic mappings added to classification prompt. 6/6 Bay Area test cases correctly mapped.

URL Deduplication

RESOLVED v1.2.2

Some articles appear through multiple RSS feeds or with variant URLs, causing duplicate classifications. Resolved in v1.2.2 with MarketScreener domain-level normalization and tracking parameter stripping. Full cross-domain deduplication (handling additional aggregator domains) remains a future improvement target.

U.S. Markets Only

V2

The current index covers 21 U.S. markets. International expansion (UAE, United Kingdom, Singapore, and parts of Asia) is under consideration for a future version. The scoring methodology is designed to be adaptable to international markets.

Graduated Scoring Factors

PARTIAL

Six of seven scoring factors remain binary (present/absent). State Legislation was converted to a graduated three-tier model in v1.2.1. Expansion of graduated scoring to additional factors (e.g., regulatoryPosture, operator presence intensity) is under consideration for a future methodology version.

> AirIndex is committed to publishing methodology updates as the scoring system evolves. Version history is maintained at airindex.io/methodology. Questions regarding methodology or data sourcing may be directed to **hello@airindex.io**.